Experiment 14

Cracking the Password Using Hydra

Aim:

- To learn how to use Hydra for password auditing by performing a controlled password-cracking experiment on an authorized test system.
- To understand how dictionary-based password attacks work and the importance of using strong passwords.

Procedure:

1. Open a Linux terminal (such as Kali Linux).
2. Ensure Hydra is installed on the system.
3. Create or obtain a test username and a password wordlist for the authorized target.
4. Identify the target IP address or hostname and the service to be tested (e.g., SSH, FTP, HTTP Login).
5. Execute the Hydra command with the appropriate options (username, password list, target, and service).
6. Monitor the progress of the password audit.
7. If a matching password is found, record the result for analysis.
8. Stop the test after completion and document the observations.
9. Analyze the importance of strong passwords and account lockout mechanisms to prevent brute-force attacks.

OUTPUT

```
kali@kali: ~
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ echo "admin" > username.txt

┌──(kali㉿kali)-[~]
└─$ echo "password123" > password.txt

┌──(kali㉿kali)-[~]
└─$ nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 12:38 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000020s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds

┌──(kali㉿kali)-[~]
└─$ hydra -L username.txt -P password.txt ssh://localhost
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizati
ons, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-04 12:38:56
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t
 4
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ssh://localhost:22/
[ERROR] could not connect to ssh://127.0.0.1:22 - Connection refused
```